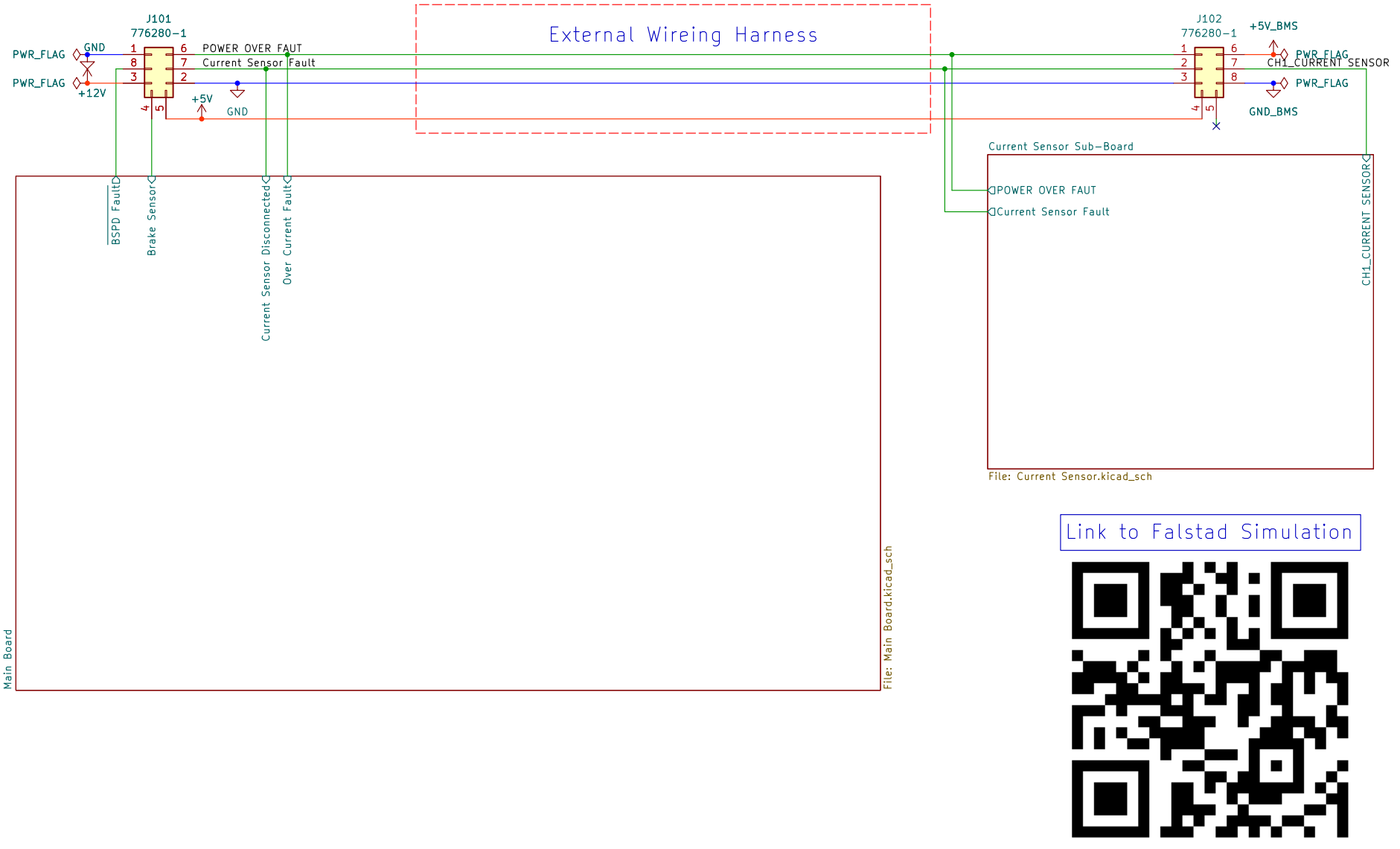


Link to Falstad Simulation

Current Sensor Test Wires -- Under Test, flow 1A over cable from Constant Current Power Source

J103
Conn_01x01_Socket
EV.5.2.5 Must Be rated for max Tractive System voltage

J104
Conn_01x01_Socket
Crimp connector to screw

15 Loops Around sensor

If 15 loops can not be made adjust current in wire

Batt. Nominal: 330V
Trip Power: 5kW
Trip Current: P = IV ⇒ I = P/V
I = 5000W/330V = 15.15A

Current per loop = I / n_{loops}
I_{test} = 15.15A/15 ≈ 1A

Dylan Shah
Carter Kortan

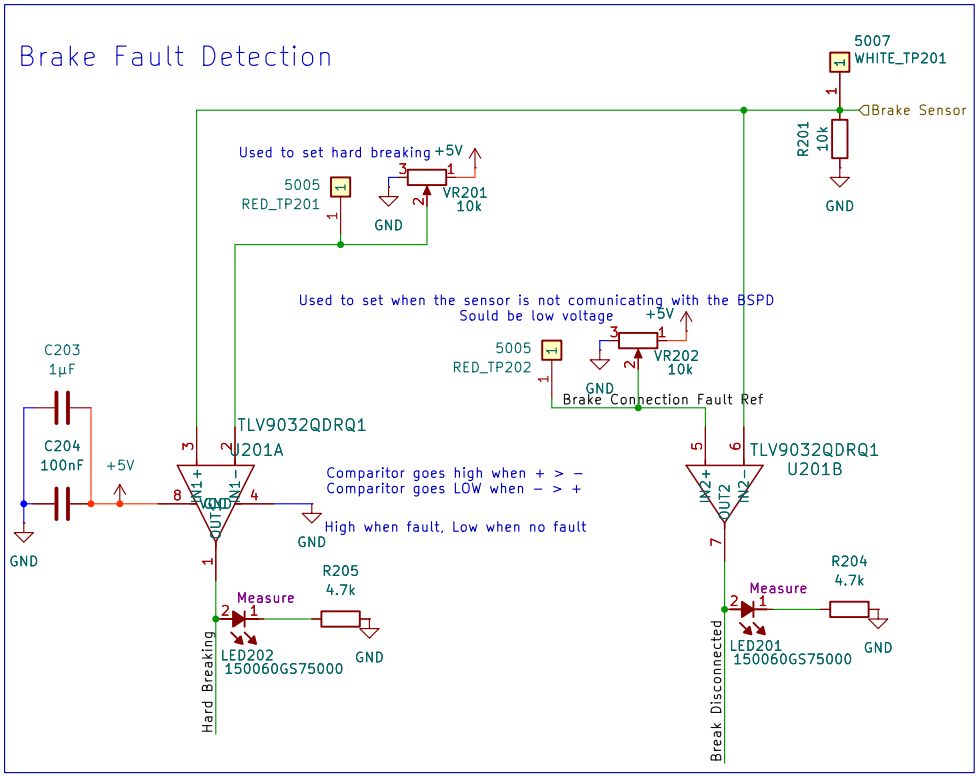
Sun Devil Motorsports

Sheet: /
File: sdm27ev-bspd-v2.kicad_sch

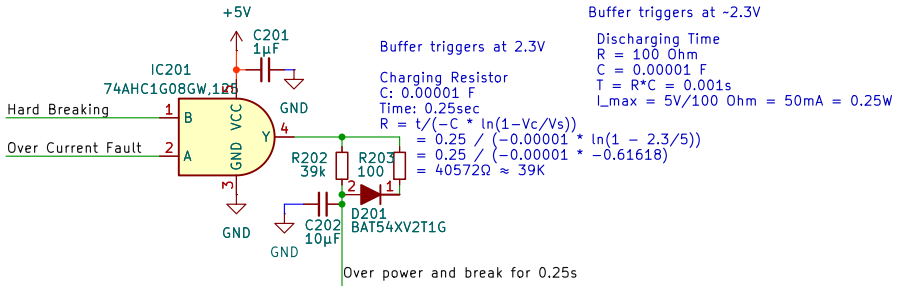
Title: BSPD

Size: USLetter	Date: 2026-10-24	**Rev: 2**
KiCad E.D.A. 10.0.8		Id: 1/3

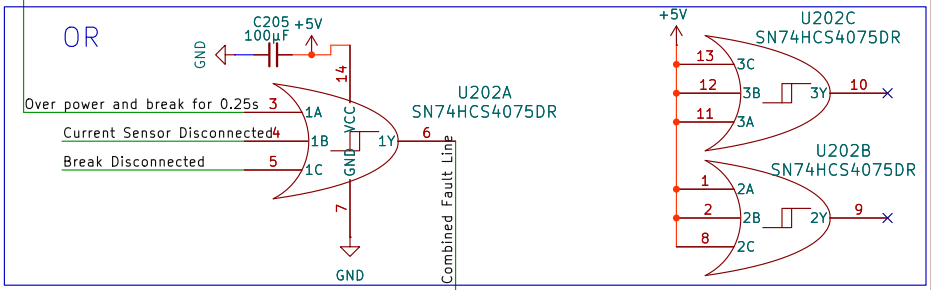
Brake Fault Detection



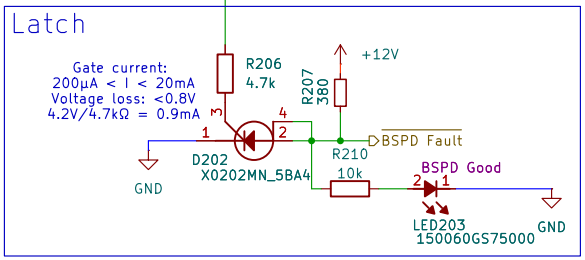
Test Hard Breaking & Over Current for > 0.25s



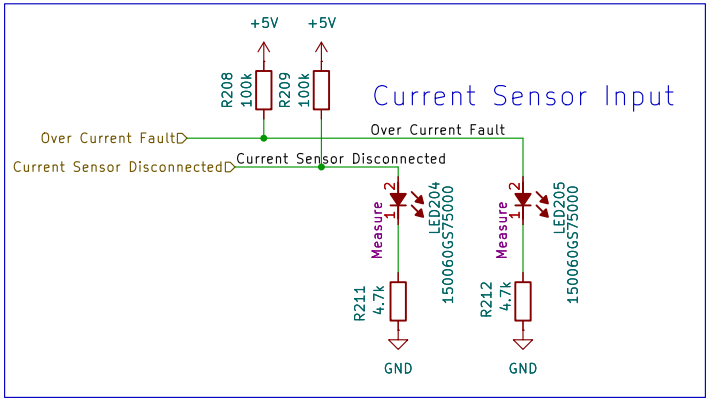
OR



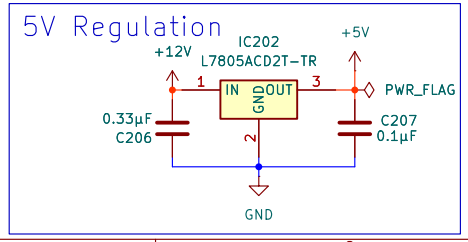
Latch



Current Sensor Input



5V Regulation



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Carter Kortan

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Sheet: /Main Board/
File: Main Board.kicad_sch

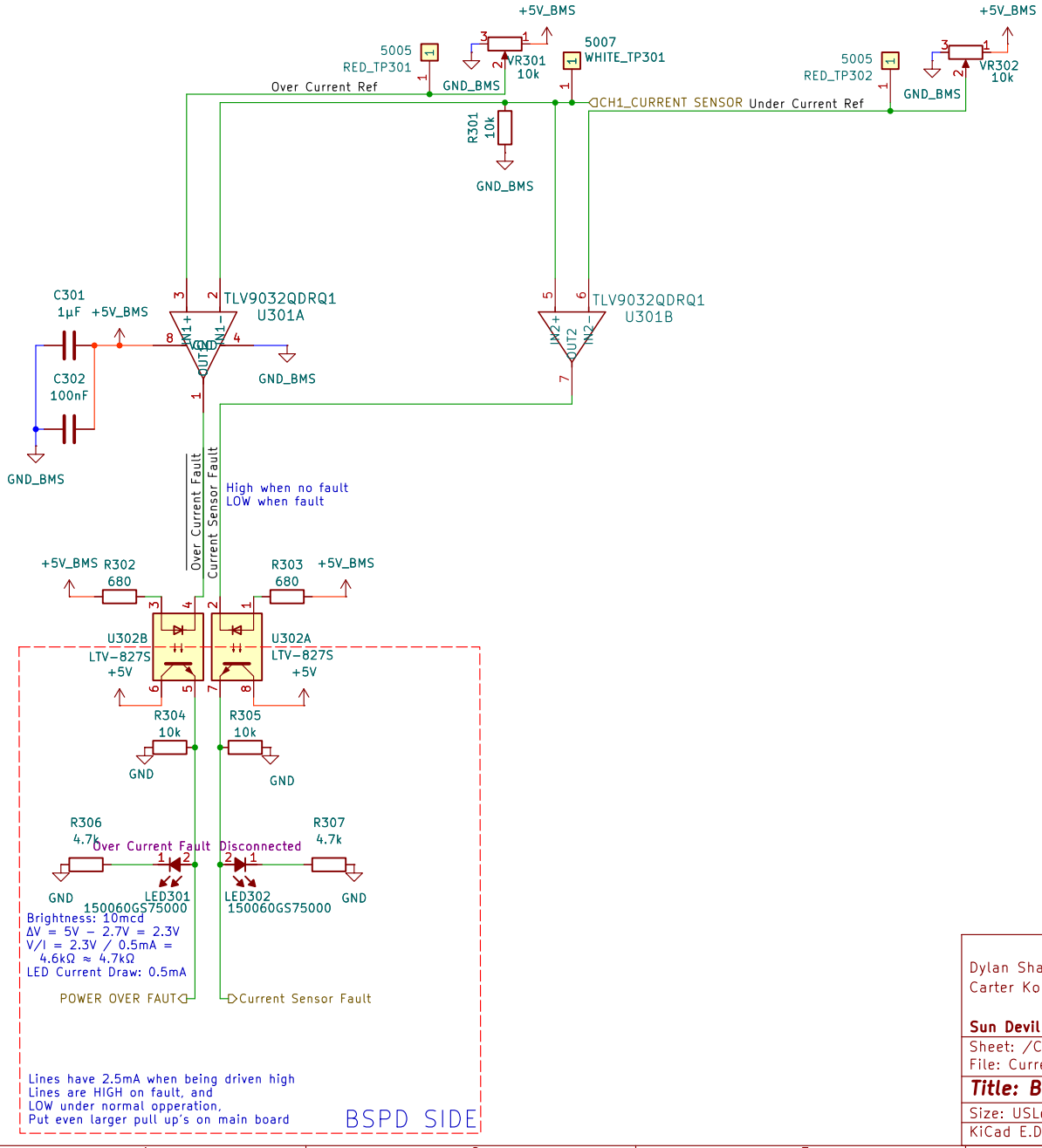
Title: BSPD Main Board

Size: USLetter	Date: 2026-10-24	**Rev: 2**
KiCad E.D.A. 10.0.8		Id: 2/3

This circutry is happning on the BMS side of the isolation between BMS and BSPD.
This is because the sensor has a max output of 1mA, and I could not find any way to isolate it without bringing BMS 5V here.
At that point, we might as well do the digital comparison beforehand, and then isolate the signal.

Used to set >5kW
Assuming this is a 330V battery, this should be ~3V

Used to set when the sensor is not comunicating with the BSPD
Sould be ~0.5V



Dylan Shah
Carter Kortan

Sun Devil Motorsports

Sheet: /Current Sensor Sub-Board/
File: Current Sensor.kicad_sch

Title: BSPD – Current Sensor Sub Board

Size: USLetter	Date: 2026-10-24	**Rev: 2**
KiCad E.D.A. 10.0.8		Id: 3/3